

PERFORMANCE WORK STATEMENT (PWS)

FOR

SUPPORT SERVICES FOR HEWLETT PACKARD ENTERPRISE HIGH PERFORMANCE COMPUTE NODE

PART 1

GENERAL INFORMATION

1.1 General: This is a non-personnel services contract to provide on-site Hewlett Packard Enterprise (HPE) High-Performance Compute (HPC) system administration support services to be provided to the Naval Oceanographic Office (NAVOCEANO or NAVO) for a period of one year and is similar to the system administration support services provided on contract N0018925P0496. The Government shall not exercise any supervision or control over the contract service providers performing the services herein

1.2 Background

1.2.1 The Naval Oceanographic Office (NAVOCEANO) collects, processes, and disseminates complex oceanographic data, forecasts, and products to Fleet users using HPC technologies including Artificial Intelligence (AI) and Machine Learning (ML) algorithms. NAVOCEANO requires specialized services for the system administration, operational support, and cybersecurity compliance of a new Hewlett Packard Enterprise (HPE) High-Performance Compute (HPC) cluster. This system is a strategic asset designed to run advanced Artificial Intelligence (AI) and Machine Learning (ML) algorithms. The architecture consists of cutting-edge technologies, including GPU-accelerated compute nodes, an HPE Slingshot high-speed interconnect, a Lustre-based parallel file system, and an image-based operating system deployment infrastructure. Due to the new technology and complexity of these components, expert contractor support is necessary to ensure the system's stability, security, and optimal performance to meet mission-critical objectives

1.3 Description of Requirement

The contractor shall provide specialized support services for system administration, operational support, and cybersecurity compliance of the HPE HPC system. The contractor shall provide all personnel, equipment, tools, materials, supervision, and other items necessary to perform the services.

1.4 Period of Performance, Place of Performance and Hours of Operation

1.4.1 The period of performance shall be 12 months after award.

1.4.2 The contractor shall perform services at NAVOCEANO. NAVOCEANO is located at Stennis Space Center, Mississippi.

1.4.3 Recognized Holidays/Government Closure: Federal law (5 U.S.C. 6103) establishes the public holidays for Federal employees. Any other day designated by federal statute, executive order, presidential proclamation and/or local determinations relating to adverse weather conditions, national emergencies, administrative closings, or similar Government directed facility closings may require office closure. During such periods of closure, contractor employees will not be allowed to work, nor will the contractor be compensated, however, offsite work is allowed, with Contracting Officer's (KO) approval, on a case-by-case basis when mutually agreed upon by the Government and the contractor.

1.4.4 Hours of Operation

The contractor is responsible for performing support services during an 8-hour workday between the hours of 0700 and 1630, Monday through Friday, except when the government facility is closed or government personnel are not in the designated office location. The contractor must maintain an adequate workforce for the uninterrupted performance of all tasks defined within this performance work statement when the Government facility is not closed. The stability and continuity of the contractor workforce are essential.

1.5 Post Award Conference/Periodic Progress Meetings

1.5.1 The Contractor shall attend any post award conference convened by the contracting activity. The contracting officer and the Contracting Officers Representative (COR) may meet periodically with the contractor to review the contractor's performance. At these meetings, the contracting officer will apprise the contractor of how the Government views the contractor's performance and the contractor will inform the Government of problems experiencing, if any. The Government or the contractor, as applicable, shall take appropriate action to resolve outstanding issues. These meetings shall be at no additional cost to the Government.

1.6 Program Management

1.6.1 The contractor shall assign a Program Manager (PM)/Contract Manager (CM) to act as the single primary point of contact with the NAVOCEANO COR. The PM/CM specifically has the responsibility of communicating with the COR on all matters pertaining to this Performance Work Statement (PWS). The PM/CM shall direct the individual contractor employee's effort through the company's internal management system; this management system shall provide project progress visibility to assure on-time completion of all requirements.

1.6.2 The contractor shall utilize standard processes based on industry best practices wherever applicable.

1.6.3 The contractor shall sponsor quarterly program management reviews to monitor contractor performance and discuss any issues that the contractor and/or NAVOCEANO are experiencing with the project referenced herein with the COR and/or Contracting Office. NAVOCEANO COR will schedule this review; submit a suggested agenda to the contractor who shall be responsible for all briefing slides and materials.

1.6.4 The contractor shall support other NAVOCEANO briefings and meetings as applicable by providing feedback and documentation/numbers concerning any efforts.

1.7 Identification of Contractor Employees: All contract personnel are required to identify themselves as such while attending meetings, answering Government telephones, answering emails and working in other situations where their contractor status is not obvious to third parties. Identification of contractor status is required to avoid creating an impression in the minds of members of the public or other Government personnel that they are Government officials. They must also mark all documents or reports produced by contractors as contractor products or that contractor participation is appropriately disclosed. All contractor personnel shall obtain and wear badges in the performance of this service

5 Personnel Education / Experience Requirements

5.1 Education:

5.1.1 All contractor personnel conducting system administration services, at a minimum, the combination of education and experience required to qualify for the Office of Personnel Management (OPM) General Schedule Qualifications Standard, IT Specialist Series 2210 or Computer Scientist series 1550. The IT Specialist Series 2210 qualifications build upon the general requirements of the 2210 series with a specific focus on the technical skills needed to manage and maintain network infrastructures, operating systems, and applications. The Computer Scientist Series 1550 qualifications Requires one of the following basic education requirements:

Bachelor's Degree in Computer Science: A bachelor's degree in computer science from an accredited institution.

Bachelor's Degree with Specific Coursework: A bachelor's degree with 30 semester hours in a combination of mathematics, statistics, and computer science. Of these 30 semester hours, at least 15 must have included a combination of statistics and mathematics, including differential and integral calculus.

All academic degrees and coursework must be from accredited or pre-accredited institutions

5.1.2 All contractor personnel conducting system administration shall have, at minimum, the combination of education and training required to qualify for the Office of Personnel Management (OPM) General Schedule Qualifications Standard, IT Specialist 2210 series at or above the GS12 level which is equivalent to four (4) years of specialized experience in the fields of IT system administration and IT network administration of HPC systems used for scientific data processing using AI/ML. More specifically LINUX based operating systems, parallel file systems, GPU-accelerated compute nodes, and high-speed network interconnects.

5.2 All contractor personnel shall have specialized knowledge of HPC systems used for scientific data processing using AI/ML. More specifically LINUX based operating systems, parallel file systems, GPU-accelerated compute nodes, and high-speed network interconnects.

PART 2 DEFINITIONS & ACRONYMS

- 2.1. **CONTRACTOR.** A supplier or vendor awarded a contract to provide specific supplies or service to the Government. The term used in this contract refers to the Prime Contractor.
- 2.2 **CONTRACTING OFFICER.** A person with authority to enter into, administer, and or terminate contracts, and make related determinations and findings on behalf of the Government. Note: The only individual who can legally bind the Government.
- 2.3 **CONTRACTING OFFICER'S REPRESENTATIVE (COR).** An employee of the U.S. Government designated and authorized in writing by the contracting officer to perform specific technical or administrative functions. Such appointment shall be in writing and shall state the scope of authority and limitations. This individual has authority to provide technical direction to the Contractor if that direction is within the scope of the contract, does not constitute a change, and has no funding implications. This individual does NOT have authority to change the terms and conditions of the contract.
- 2.4 **DEFECTIVE SERVICE.** A service output that does not meet the standard of performance associated with the Performance Work Statement.
- 2..5 **QUALITY ASSURANCE.** The Government procedures to verify the services performed by the Contractor are within acceptable standards.
- 2.6 **QUALITY CONTROL.** All necessary measures taken by the Contractor to assure that the quality of a product or service shall meet contract requirements.

PART 3 SPECIFIC TASKS

3.1 Task 1: System and Network Administration

The contractor shall provide expert system administration for the entire HPC ecosystem.

- **3.1.1 Hardware and Software Management:** Install, configure, and maintain all HPC components, including compute nodes (CPU/GPU), Slingshot interconnect fabric, and the Lustre parallel file system.
- **3.1.2 Image-Based OS Deployment:** Manage the creation, maintenance, testing, and deployment of operating system images for compute and service nodes. Ensure a stable and consistent software environment.
- **3.1.3 User and Resource Management:** Manage user accounts (create, modify, delete), access controls, enforcing security policies, and resource allocation. Implement and manage a job scheduling system (e.g., Slurm) to ensure equitable and efficient use of computational resources.
- **3.1.4 System Monitoring and Maintenance:** Proactively monitor system health, performance, and utilization. Perform routine maintenance, apply firmware updates, and manage hardware repairs or replacements in coordination with the vendor.
- **3.1.5 System Documentation:** Maintaining detailed system documentation, including configuration information, troubleshooting procedures, and security protocols.
- **3.1.6 Manage Network Architecture:** Configure and manage the HPC network infrastructure, including switches, routers, and firewalls (HPE Slingshot). Optimize network performance, investigate and resolve network connectivity issues, and ensure network security.
- **3.1.7 Storage Management:** Manage the HPC storage system, including configuring and maintaining the file system (e.g., Lustre, HPE Parallel File System Storage, GPFS). Optimize storage performance for HPE HPC workloads, implement data backup and recovery procedures. Monitor storage capacity and usage.
- **3.1.8 Software Deployment and Management:** Install, configure, and maintain HPC software applications, including compilers, libraries, and scientific applications. (e.g. HPE Cray System Management, HPE Performance Cluster Manager, Slurm, Docker). Manage software licenses. Ensuring software compatibility. Supporting users in using HPC software.
- **3.1.9 Provide Expertise:** Provide advice and guidance on best practices for HPC system administration.

3.2 Task 2: Operational Support

The contractor shall provide operational support to end-users and technical staff to maximize the utility and performance of the HPC system for AI/ML workloads.

- **3.2.1 User Support:** Provide technical support to users and serve as the primary point of contact for resolving user issues encountered on the HPC system performance. Develop user documentation and conduct user training sessions.
- **3.2.2 Performance Tuning and Optimization:** Analyze and optimize AI/ML algorithm performance on the HPC cluster. Provide expert guidance to users on code parallelization, GPU utilization, and efficient use of the Slingshot interconnect and Lustre file system.
- **3.2.3 Software Stack Management:** Install, configure, and maintain a robust software stack for AI/ML development, including compilers, scientific libraries, machine learning frameworks (e.g., TensorFlow, PyTorch), and containerization technologies (e.g., Docker, Singularity/Apptainer).
- **3.2.4 Knowledge Transfer:** Develop and deliver training sessions and comprehensive documentation for government personnel on the administration and use of the HPC system. The goal is to enable a smooth transition to full government ownership and support at the conclusion of the contract period.

3.3 Task 3: Cybersecurity Compliance

The contractor shall ensure the HPC system meets all applicable Department of War (DoW) and DoD cybersecurity standards.

- **3.3.1 System Hardening:** Implement and maintain system configurations in accordance with Security Technical Implementation Guides (STIGs) and other mandated security controls.
 - **3.3.1.1** Quarterly, at each STIG cycle release by DISA, evaluate cybersecurity compliance of the HPC system against the most recent DISA STIG and create updated security checklists.
 - **3.3.1.2** Implement cyber security requirements due to CTO, directives or DISA STIG updates.
 - **3.3.1.3** Maintain a system administration guide with details on how to remediate each STIG item and step-by-step directions on manual implementation details for those STIG items that must be implemented manually
 - **3.3.1.4** Create and/or modify scripts and documentation supporting STIG compliance.
- **3.3.2 Vulnerability Management:** Conduct regular vulnerability scanning of the HPC environment. Remediate identified vulnerabilities through patching, configuration changes, or other mitigation measures in a timely manner.
- **3.3.3 Security Monitoring and Incident Response:** Implement and monitor security logging and auditing capabilities. Respond to security alerts, investigate potential incidents, and provide detailed reports as required.
- **3.3.4 Compliance Documentation:** Develop and maintain all necessary documentation to support the system's Assessment and Authorization (A&A) process and achieve an Authority to Operate (ATO).
- **3.3.5 Security Patches/Upgrades:** Applying security patches and updates promptly in accordance with NAVOCEANO and DoN policies.
- **3.3.6 Cybersecurity Compliance:** Ensure compliance with all applicable cybersecurity requirements and best practices, Defense Information Systems Agency (DISA) Security Technical Implementation Guides (STIG), Department of War Zero Trust Directives, Department of Navy Risk Management Framework, NIST 800-53 and agency-specific security policies.

3.4 Task 4: Knowledge Transfer and Training

- **3.4.1 On-the-Job Training:** Providing training and mentorship to NAVO's IT staff in critical areas such as security best practices, system configuration, troubleshooting, and maintenance procedures.
- **3.4.2 Deliver Formal Training:** Provide formal training sessions on HPC system administration topics.
- **3.4.3 Knowledge Base:** Create documentation and knowledge base articles to support ongoing training and knowledge sharing.
- **3.4.4 Knowledge Transfer:** Facilitating knowledge transfer with the intention that NAVO's staff can independently manage the system after the one-year support period.

PART 4

GOVERNMENT FURNISHED PROPERTY / EQUIPMENT / RESOURCES

4.1 General: Except for those items specifically stated in paragraph 4.2, 4.3 and 4.4 the contractor shall furnish everything required to perform this PWS.

4.2

4.2.1 Equipment: Contractor personnel will have access to computer systems in specified NAVOCEANO office space to perform tasks outline in the PWS.

4.2.2 Software: N/A. The computer systems provided for contractor will have the software required to perform the tasks in the PWS.

4.3 Facilities: The Government will furnish the necessary workspace for the contractor staff to provide the support outlined in this PWS to include desk space, telephones, computers and other items necessary to maintain an office environment.

4.3.1 The contractor shall return to NAVOCEANO all materials furnished to the contractor and any materials purchased with the Government funds in support of this effort as directed, or upon completion of this project.

4.3.2 The contractor shall only use the data and software provided to the contractor under the terms of fulfilling their contractual obligations. The contractor shall transition all data and software back to NAVOCEANO upon project completion or contract performance period. Upon delivery, the contractor shall return to NAVO and remove from contractor holdings all original and derived data and products.

4.4 Utilities. All utilities in the facility will be available for the contractor's use in performance of duties outlined in this PWS. The Contractor shall instruct employees in utilities conservation practices. The contractor shall be responsible for operating under conditions that preclude the waste of utilities.

4.5 Standard Operating Procedures (SOPs) and System Access

4.5.1 Standard Operating Procedures (SOPs) and Guidance for system, network administration, and NAVOCEANO processes will be provided as needed.

4.5.3 NAVOCEANO will provide system access within 30 business days of the contractor completing all required security training, security application paperwork, and system access requests.

4.5.4 3 NAVOCEANO will grant access necessary for the completion of this effort upon receipt of required scientific computer accounts.

4.5.6 NAVOCEANO personnel will provide tasking familiarization. Familiarization for new contractor employees shall consist of SOP review and software orientation. The duration of this familiarization shall be at the discretion of NAVOCEANO and shall not exceed five business days

PART 5

SECURITY REQUIREMENTS / INFORMATION TECHNOLOGY (IT) SYSTEMS ACCESS

The overall classification of this contract is CUI (Controlled Unclassified Information). Contractor does not require access to classification information in the performance of this contract. All contract personnel shall comply with the directives and policies established in 2 CFR Part 117 "National Industrial Security Program Operating Manual", DoD Instruction 5200.48 (Controlled Unclassified Information) and SECNAVINST 5510.36B "Department of the Navy (DoN) Information Security Program." Upon award of the contract (or within 30 days of contract award) the Contract Company will provide the COR with contractor information. COR will provide the Security Office the contractor information to begin the required clearance investigations or to verify the contractors have the required investigation for the contract.

PART 6

PERFORMANCE REQUIREMENTS SUMMARY AND QUALITY CONTROL

6.1 Performance Monitoring

6.1.1 This PWS not only discusses the work to be performed but also contains performance standards and acceptable performance levels (APL). Performance standards are "tools" the Government uses to measure level of performance. A standard is an optimum performance level against which actual performance is measured or evaluated. For example, in the statement "complete a service call in 24 hours" the standard is 24 hours. The number of days taken to finish this task is the performance indicator, which determines if the work performed was below, met, or exceeded the standard. Developing accurate performance standards is crucial to ensuring that the actual performance received, regardless of the service provider, meets the requirements of the Statement

of Work. This PWS includes performance standards for each of the Functional Areas of the PWS. The performance standards provided with this PWS seek to state the characteristics of properly completed outputs. Each performance standard contains standards for both quality and timeliness.

6.1.2 The COR will inspect for compliance with Contract terms and conditions throughout the Contract period. The Government will monitor the Contractor's performance by performing surveillance using the PRS. The PRS is used primarily to determine if the Contractor is performing all required outputs. The PRS also indicates for contractors the importance of each required service.

6.1.3 DFARS 222.17 mandates including surveillance for ensuring compliance with Combatting Trafficking in Persons (CTIP). The COR will monitor the contractor's performance regarding trafficking in persons through the checklist for auditing compliance with CTIP policy found at DFARS PGI 222.17. Such non-compliance with FAR clause 52.222-50, Combating Trafficking in Persons, will be brought to the immediate attention of the contracting officer. See the DoD CTIP website (<https://ctip.defense.gov>) for further information

6.2 Quality Control/Assurance

6.2.1 The Government shall rely on the contractors' existing quality assurance systems for the commercial item being acquired as per customary industry practices. The quality control plan shall ensure configuration, maintenance and repair services are performed in accordance with manufacturer's recommendations and generally accepted commercial practices. The plan shall identify procedures to identify and prevent defective services from recurring.

6.2.2 The contractor shall provide a copy of their quality control program to the Government within 30 days of award.

Desired Outcomes	PWS Paragraph	Required Service	Performance Standard/Measurement	Monitoring Method	Incentives/Disincentives for Meeting or Not Meeting the Acceptable Quality Level
Quality of Service	Adhere to requirements specified in paragraph 4.0 of the PWS	100% accuracy rate for actions completed	Quality/Timeliness	Random Inspections/ Customer Surveys or feedback/ Review of work Requests	(a) Satisfactory or above ratings on past performance. (b) Any contractor deficiency in operation or delay of installation of equipment must provide negotiated consideration
Prepare and submit reports	Prepare monthly reports of deliverables provided	98% (2% error) for every report completed	Quality/Timeliness	100% Inspection	(a) Satisfactory or above ratings on past performance. (b) Any contractor deficiency must provide negotiated consideration
Prepare and submit documentation	4.0 Deliverables	98% (2% error) for every document completed	Quality/Timeliness	100% Inspection	(a) Satisfactory or above ratings on past performance. (b) Any contractor deficiency must provide negotiated consideration

PART 7
DELIVERABLES SCHEDULE

4.0 Deliverables

The Contractor shall provide the following deliverables:

- **System Administration & Operations Plan:** Plan outlining details of system administration duties.
- **Monthly Status Reports:** Detailed reports summarizing the Contractor's activities, accomplishments, challenges, and recommendations for improvement. The report should include metrics on system uptime, performance, and security incidents.
- **System Documentation:** Comprehensive documentation of the HPC system configuration, including hardware and software inventory, network diagrams, security protocols, and troubleshooting procedures.
- **Training Materials:** Documentation, presentations, and other materials used in the delivery of training sessions.
- **Knowledge Base Articles:** Articles documenting solutions to common problems and best practices for HPC system administration.
- **Security Assessment Reports:** Reports detailing the results of security assessments and vulnerability scans, along with recommendations for remediation.
- **Incident Reports:** Reports documenting security incidents, including the cause of the incident, the actions taken to remediate the incident, and recommendations for preventing future incidents.
- **Transition Plan:** A detailed plan outlining the process for transitioning the HPC system administration responsibilities to NAVO's IT staff.

All technical and cybersecurity related deliverables must comply with all applicable DoD and NAVOCEANO security configuration or implementation guides. This includes but is not limited to Defense Information Systems Agency (DISA) Secure Technical Implementation Guides (STIGs) and

Reference: DODI 8500.2, IA

Control: ECSC-1

Subject: Security Configuration Compliance

Requirement: For Enclaves and ATS applications, all DoD security configuration or implementation guides have been applied.

Title	Format Required	Frequency	Due date	Mission Impact
System Administration & Operations Plan	Microsoft (MS) Word Doc	Updated if plan changes	14 days after contract award	Medium
Monthly Status Report (MSR)	MS Word	Monthly	Due on the 5th business day following the close of the calendar month.	Low
Security Assessment Reports	MS Word	As needed	After cybersecurity Assessment scans	High
Incident Reports	MS Word	As needed	As specified by ISSM	High
System Documentation	MS Word	Initial baseline, updates as configuration changes	In each release package	High

STIG Checklist	DISA STIG viewer format .ckl	Quarterly	As needed by ISSM	High
Training Materials/knowledge base Articles	Microsoft Word	As Completed	As identified; at least 60 days before end of performance	High
Transition Plan	Microsoft Word		at least 60 days before end of performance	High

PART 8 APPLICABLE PUBLICATIONS

PART 9

LIST OF ATTACHMENTS

There are no attachments associated with this PWS.